



Introducing Samba 4

Now, Even More Awesomeness

Samba 4 has been in development for a long time but an official first release is imminent, the developers say. Its biggest feature is Active Directory Server support, which removes the last hurdle in a pure-Linux server set-up, with only Windows PCs and Macs as clients. In this 'FOSS for Windows' special issue of *LFY*, let's explore how to set up Active Directory Server, so that you can finally phase out those pesky Windows Servers while keeping Windows desktops and Macs intact.

If you've ever used Windows and Linux together on a network, I'm betting my entire wallet (not that there's anything in it at the moment, but that's beside the point) that you've used Samba. If you haven't (or have, without knowing it), Samba is the software that lets you access Windows network shares from Linux. Not just that—Samba's built-in first-class server lets you create Windows-accessible shares on your Linux box.

Samba 3 can also host a Windows NT 4-compatible Domain set-up, and can act as a PDC (primary domain controller) as well as a BDC (back-up domain controller). But who uses NT 4 Domains any more? Ever since Windows 2000 Server was released at the end of 1999, everyone's been using Active Directory (AD) to organise clients and hold user and computer data. And why not? AD is based on industry standards—Kerberos 5 for authentication, LDAP for the resource database, and all of it glued together with DNS SRV records. So why did it take 13 years for Samba to finally support AD?

Proprietary extensions to LDAP and Kerberos 5, that's why. But those extensions have now become open standards, and the AD LDAP schema has been reverse-engineered, so Samba can finally start acting as an Active Directory Server (ADS). You can now convert all your Windows Servers to Linux, without changing a line of code (Samba 4 can now do AD, and Mono can do ASP.NET, which are actually the only two reasons people use Windows Servers anyway).

Getting started

First of all, Samba 4 is still in development, but also stable enough for Canonical to include it in the official repositories for Ubuntu 12.04 LTS (Long-Term Support release). With the light testing I've been doing at home, it works pretty well, so if you're going to think about rolling this out across your corporate infrastructure, you could try adding a couple of Samba boxes as backup AD servers, and see how they handle the load.

You've probably realised by now that Samba 4 is currently available only on Ubuntu 12.04. At least, a sufficiently updated non-alpha version that works—even though the packaging is broken (which we can fix, so there's no need to worry). I have a hunch this package will never be updated, but I also believe that once official releases are out, there'll be a PPA providing packages for this.

Before we start, we need a box that has Ubuntu 12.04 LTS installed. Samba 3 should NOT be installed (the configuration isn't compatible, so if it has already been installed, remember your configuration and purge all Samba 3 packages).

Setting up Samba 4

I'm assuming your DHCP server is set up and working fine. The DHCP server may be on a different machine, but you must be able to edit your DHCP configuration. More appropriately, you should be able to make your DHCP server point to custom DNS servers, so if you're doing this at home